Real-time fraud detection in instant payment systems

"What research exists on real-time fraud detection for instant payment systems such as UPI, FedNow, Faster Payments or SEPA Instant? What are the latency and accuracy challenges?" *Generated by LeapSpace*

Real-time fraud detection for instant payment systems is being studied mainly as a **streaming, low-latency classification problem** on UPI and similar rails, with researchers increasingly combining **gradient-boosting, sequence models, deep learning, graph neural networks, federated learning, and explainability** to cope with fast transaction flows, class imbalance, and evolving fraud patterns [135] [136] [137] [138] [139] [140].

What the research looks like

Research direction	What it tries to do	Representative evidence
Classical ML for UPI fraud	Use tabular features such as amount, device change, frequency, geolocation, and behavior to flag fraud quickly [136] [141] [142]	Random Forest reached **96.2% accuracy** with low-latency simulation on UPI [136]; XGBoost in a UPI fraud framework reached **F1 0.93** and **ROC-AUC 0.94** [142].
Deep learning / sequence models	Model transaction order and temporal behavior in real time [137] [143] [144]	A Bi-LSTM UPI/wallet model reported **91.8% accuracy**, **F1 90.5**, **PR-AUC 0.94**, and **95 ms** inference per transaction [137]. A hybrid Bi-LSTM + boosting system reported **98.99% test accuracy** and **0.9997 ROC-AUC** [143].
Graph-based methods	Capture mule rings and linked fraudulent activity across transactions and accounts [139] [145]	A UPI GNN system used **GraphSAGE + LightGBM**, achieved **sub-millisecond inference**, **30-second graph updates**, and kept false positives below **5%** at an operational threshold [139]. A distilled graph-transformer-to-attention model reduced inference time and model size on UPI data [145].
Federated / privacy-preserving systems	Train across institutions or preserve privacy while still detecting fraud [135] [140] [146]	A UPI framework combined **federated learning** and **XAI** for privacy-preserving, fast deployment [135]. A federated fraud model showed latency rising with more clients, but still stayed below a **300 ms** operational threshold at 50 institutions [140].

Research direction	What it tries to do	Representative evidence
Hybrid and defensive designs	Combine multiple models, add OTP checks, or harden against adversarial attacks [138] [147] [148] [149] [150] [151]	FraudGuard reported **>97% overall accuracy** with lower false negatives by routing uncertain cases to a contextual model [147]. UPI-CIR improved adversarial robustness from **F1 0.04** under attack to **F1 0.82** [149].

Latency challenges

The biggest operational challenge is that instant payment systems demand **very short decision times** while transaction volumes are high and patterns change quickly [139] [140] [152] [153] [154]. Several papers explicitly state that production fraud systems need processing in the **millisecond range**, with one source noting a practical target of **under 50 ms** for real-time fraud inference [152]. UPI-specific work shows the tension clearly: a deep learning model took **95 ms per transaction** [137], a boosted UPI system reported **45 ms delay per transaction** [138], and a Kafka-based real-time banking system reported **42 ms end-to-end latency** [153]. Graph-based systems can also be fast, but scaling graph updates remains a concern; one UPI GNN system reported **sub-millisecond inference** but still needed **30-second graph updates** [139]. More broadly, federated systems add communication and synchronization overhead as more institutions join, even when total latency remains under an operational limit [140].

Accuracy challenges

Accuracy is hard to interpret because real-time payment fraud is **highly imbalanced** and fraudsters adapt to detection rules [138] [139] [141] [148]. Several studies report strong headline accuracy, but they also warn that **false positives** and **false negatives** matter more than raw accuracy in live payment flows [136] [138] [142] [146]. One UPI GNN paper says conventional fraud systems suffered **20-30% false positive rates**, while its graph-based model kept false positives below **5%** [139]. Another study reported a **1.2% false positive rate** alongside **98.7% accuracy** and **98.5% recall** [138]. Some systems emphasize balanced metrics instead of accuracy alone: Bi-LSTM achieved **PR-AUC 0.94** on a skewed dataset with **7.5% fraud** [137], and a federated fraud model improved privacy but faced trade-offs between performance and communication cost [140].

Key papers to start with

- **Real Time UPI Fraud Detection Using GNNs** — graph-based real-time detection with mule-ring structure, sub-millisecond inference, and low false positives [139]
- **A Holistic Framework for Realtime UPI Fraud Detection** — compares model families and explicitly treats prediction latency as a first-class metric [135]

- **Machine Learning for Safeguarding of UPI Scams** — Random Forest on UPI fraud with **96.2% accuracy** and low-latency simulation [136]
- **AI-Powered Fraud Detection in UPI and Digital Wallet Transactions Using Deep Learning Models** — Bi-LSTM with **91.8% accuracy**, **PR-AUC 0.94**, **95 ms** inference [137]
- **A Deep Hybrid Learning Approach for Fraud Detection in Digital Payment Systems Using Bi-LSTM and Ensemble Boosting** — strong accuracy/ROC-AUC in UPI-like payment data [143]
- **FraudGuard: A Selective Confidence-Based Ensemble Predictive Machine Learning and Encoder-Based Language Models for Low-Latency Financial Fraud Detection** — hybrid risk scoring and contextual escalation [147]
- **A Constraint-Aware Framework for Adversarial Robustness in UPI Fraud Detection** — shows how attack-aware training can recover performance under realistic adversarial constraints [149]

Bottom line: the field is moving from generic fraud classifiers to **real-time, latency-aware systems** that combine tabular features, sequence modeling, graph structure, and privacy-preserving deployment; the main unsolved problems are **keeping inference and graph updates fast enough** and **maintaining precision/recall under extreme imbalance and adversarial adaptation** [135] [137] [139] [140] [149].

Confidence level: High — multiple independent studies across UPI and broader instant-payment settings consistently report the same latency constraints, imbalance issues, and the value of hybrid/graph-based approaches, with several papers providing concrete performance numbers [135] [136] [137] [138] [139] [140].

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